

Tutorial_6 Answer Key

1. (a) $|I| = 221.46 \text{ A}$
(b) 0.66 lag
(c) capacitor should be providing 28.5 kVAR and its capacitance value will be 1.57 mF
(d) $|I| = 153.5 \text{ A}$
(e) yes, introducing the capacitor improves the power factor and reduces the overall demand of current
2. (a) $V_{20\Omega} = 10 \text{ V}$
 $V_{coil} = 400.5 \angle 87.14^\circ \text{ V}$
 $V_C = 400 \angle -90^\circ \text{ V}$

(b) 0.5 A (c) 15 W
3. $f_C = 918.11 \text{ Hz}$, $|V_C| = 173.35 \text{ V}$
4. $L = 39.6 \mu\text{H}$
5. (a) $C = 159.3 \text{ pF}$ and $|I_{resonance}| = 5 \text{ mA}$
(b) $C = 176.93 \text{ pF}$
6. $|I_{maximum rms}| = 2 \text{ A}$
 $f_r = 7957.74 \text{ Hz}$
 $|V_{L,rms}| = |V_{C,rms}| = 400 \text{ V}$
7. $C = 989.46 \text{ pF}$
Bandwidth = 100.5 rad/s or 15.995 kHz
8. (a) $C = 0.625 \mu\text{F}$
(b) $\omega_1 = 7920 \text{ rad/s}$ $\omega_2 = 8080 \text{ rad/s}$ $B.W. = 160 \text{ rad/s}$
(c) P_{avg} at resonant frequency = 2500 W
 P_{avg} at cut – off frequencies (ω_1, ω_2) = 1250 W
9. $C = 0.543 \text{ pF}$ to 0.818 pF